ICT2217 Network Security

Lab Exercise 3.2: AAA Part 1 - Authentication

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Configure authentication using local login
- Configure authentication using centralised TACACS+ server
- Configure authentication using centralised RADIUS server

Required Hardware

- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
  - USB-to-RJ45 console cables
  - Ethernet cables
- 3 x Laptops

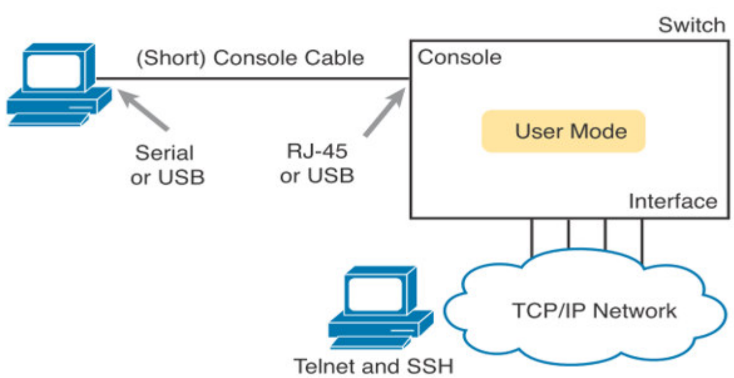
Required Software

- Tera Term https://teratermproject.github.io/index-en.html
- Kali Linux Live Boot on USB drive
- TACACS.net 2.1.2 Demo version for Windows https://www.softpedia.com/get/Internet/Servers/Other-Servers/TACACS-net.shtml
- FreeRADIUS 3.2.3 for Kali Linux

PART 1: Configuring Authentication using Local Login

1.1 Connect 1 PC to a network device, say Cisco Catalyst 9200L switch as shown:

- Using a console cable to connect to the console port
- Using an Ethernet cable to connect to one of the switch ports



Note: This lab exercise is based on accessing network devices remotely using in-band management over main production network. If OoB management LAN is setup, then the PC will be connected to the management port of the switch.

1.2 In order to be able to manage the switch remotely, first use the console connection to configure IP address to the switch as shown:

```
switch(config)# int vlan vlan-id
switch(config-if)# ip address ip-address subnet-mask
switch(config-if)# exit
switch(config)# ip default-gateway ip-address
```

Note: Ensure that the switch port is in the same VLAN as the interface VLAN. If OoB management LAN is setup, then will need to configure IP address to the management port if available.

1.3 Enable the switch to support SSH as shown:

```
switch(config)# hostname hostname
SW1(config)# ip domain name domain-name
SW1(config)# crypto key generate rsa modulus bit-size
SW1(config)# ip ssh-version 2
```

1.4 Configure the switch with local authentication as discussed in Lab3.2Notes pg 3 – 8.

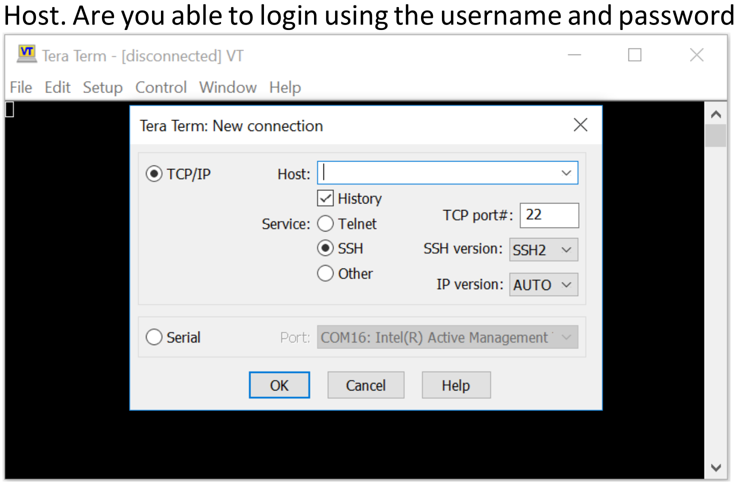
WARNING: Do NOT save your configuration to the running-config so that if you make any mistake and are unable to get authentication to work, you are able to simply reload or power down/up the network device to try again.

1.5 When done, enter 'exit' (multiple times) until you are disconnected.

1.6 Enter into CLI user mode using the console connection again. Do you need to login now? Are you able to login using the username and password that you've configured?

1.7 Configure the PC with suitable IP address so that it can ping the switch.

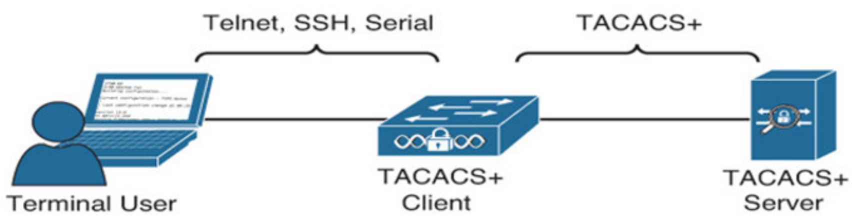
1.8 Now, try connecting via SSH on Tera Term using the IP address of the switch as the Host. Are you able to login using the username and password that you've configured?



1.9 If you are not able to login (likely due to mistakes in configuration), the only way is to reload and try re-configuring again. (That's why you are advised NOT to save your configuration.)

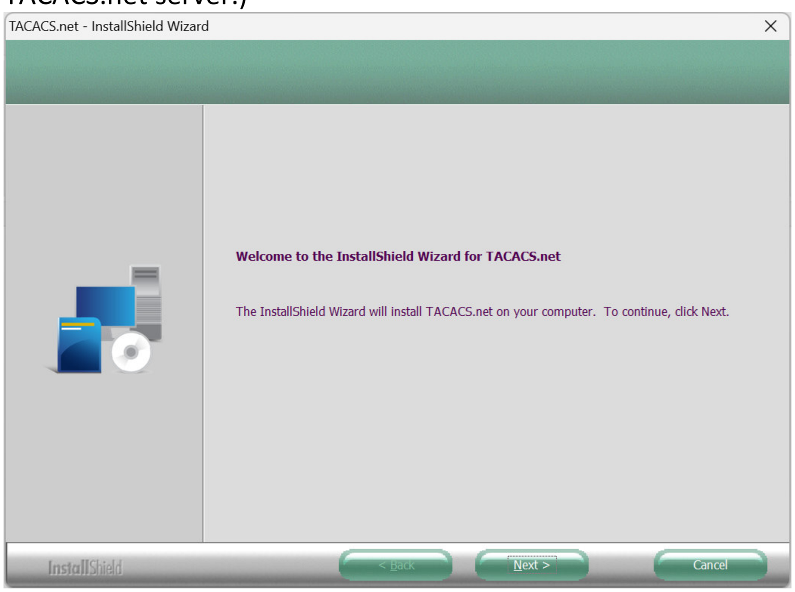
PART 2: Configuring Authentication using Centralised TACACS+ Server

2.1 Extend the network topology in Part 1 by connecting another PC to the switch port to function as a TACACS+ server as shown:

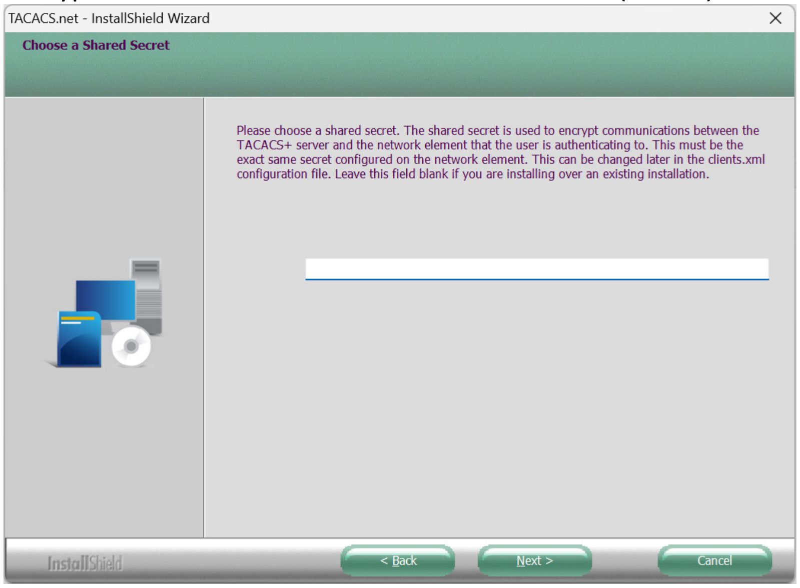


2.2 Boot up the TACACS+ PC in Windows and configure it with suitable IP address so that it can ping the switch also.

2.3 Download and install TACACS.net if necessary. (Lab PCs may already be installed with TACACS.net server.)



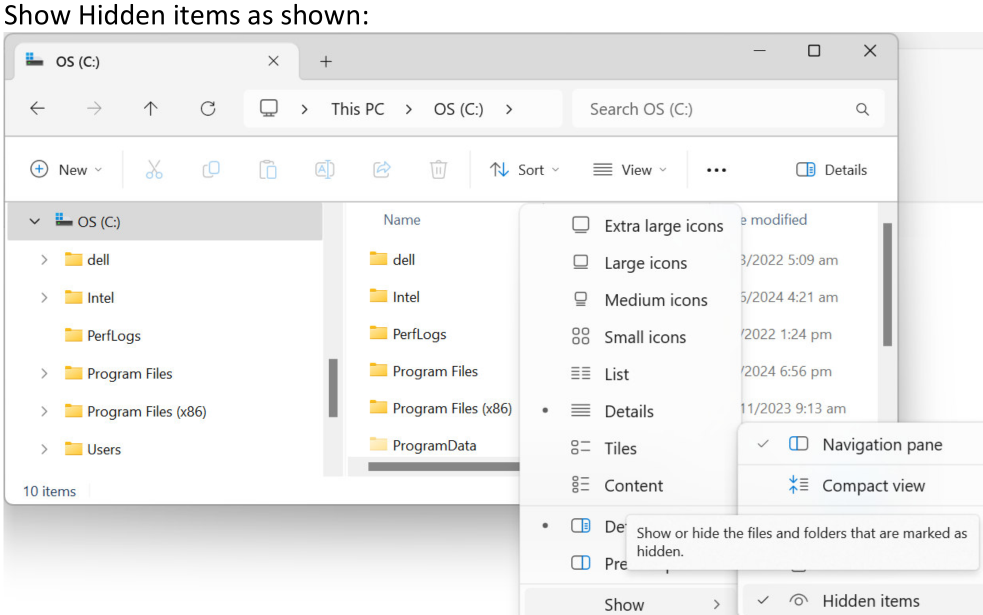
2.4 When prompted during installation, enter a shared secret password to be used to encrypt communications between TACACS+ client (switch) and TACACS+ server (PC).

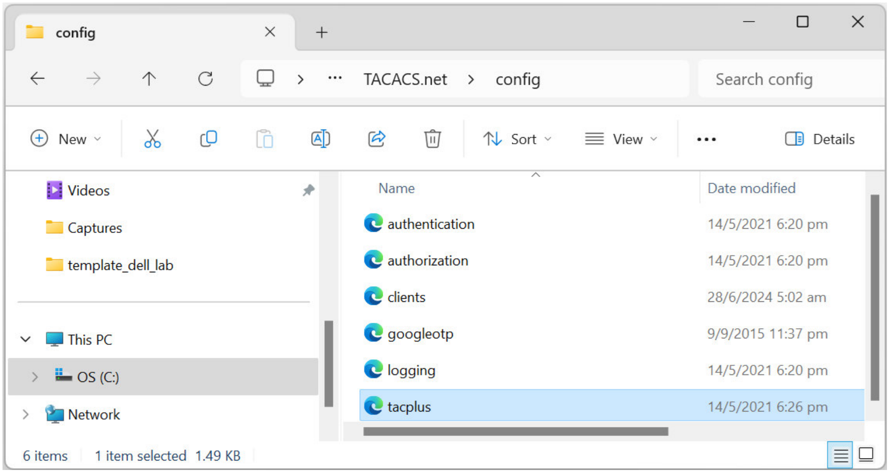


2.5 If installing using Lab PCs in Windows 11, the installed configuration files should be found at:

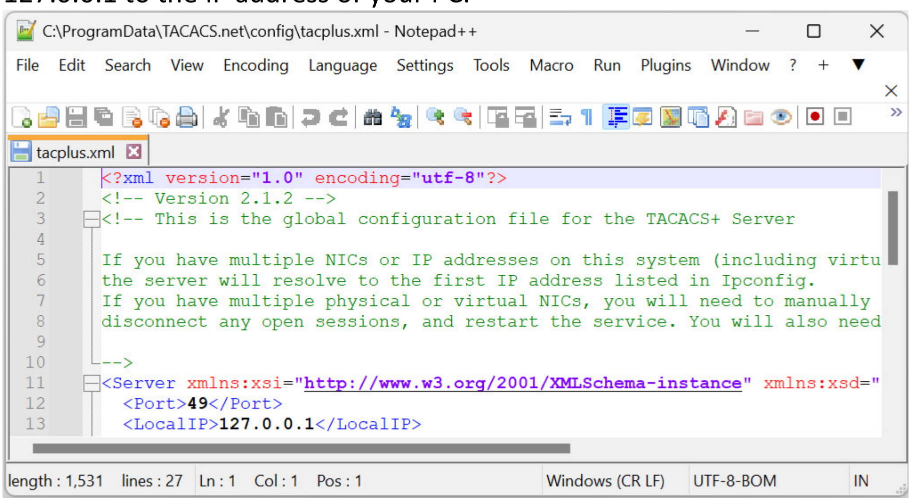
C:\ProgramData\TACACS.net\config

Note: To find the folder, you may need to click the View drop-down menu to select Show Hidden items as shown:

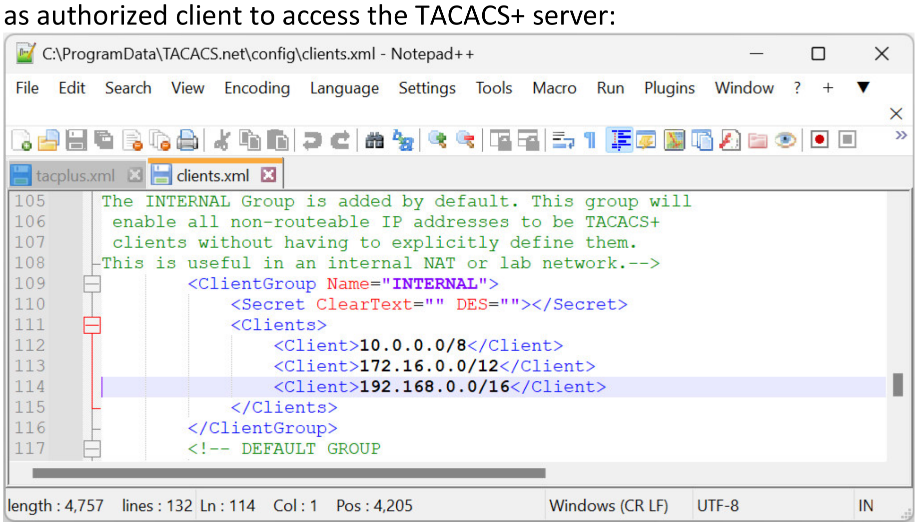




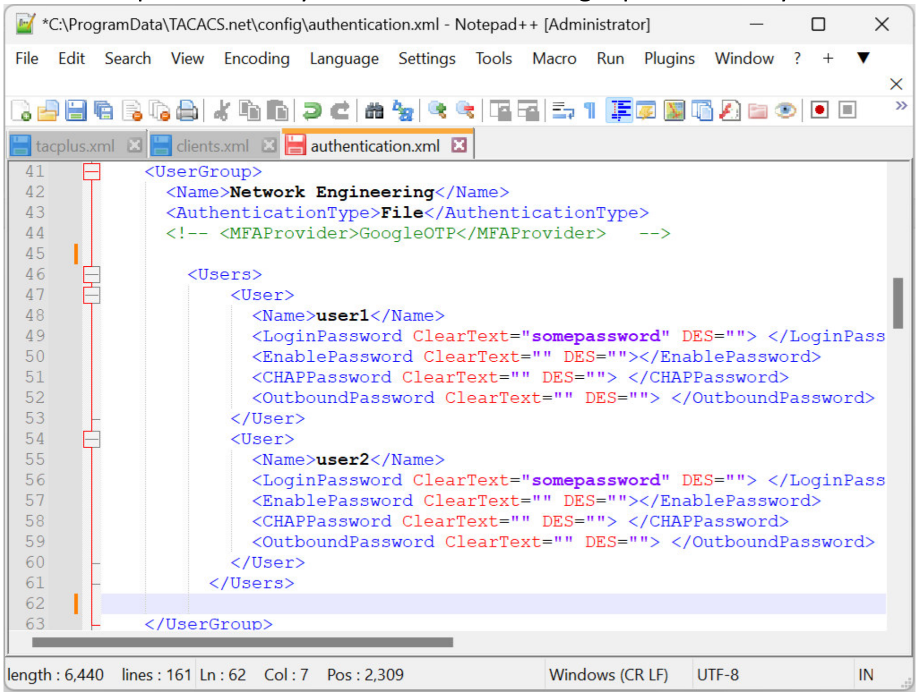
2.6 By default, the TACACS+ server will run at IP address 127.0.0.1 and TCP port number 49. Start a text editor, e.g. Notepad++, to modify the **tacplus.xml** file to replace 127.0.0.1 to the IP address of your PC:



2.7 If necessary, edit the **clients.xml** file to ensure that your switch IP address is included as authorized client to access the TACACS+ server:

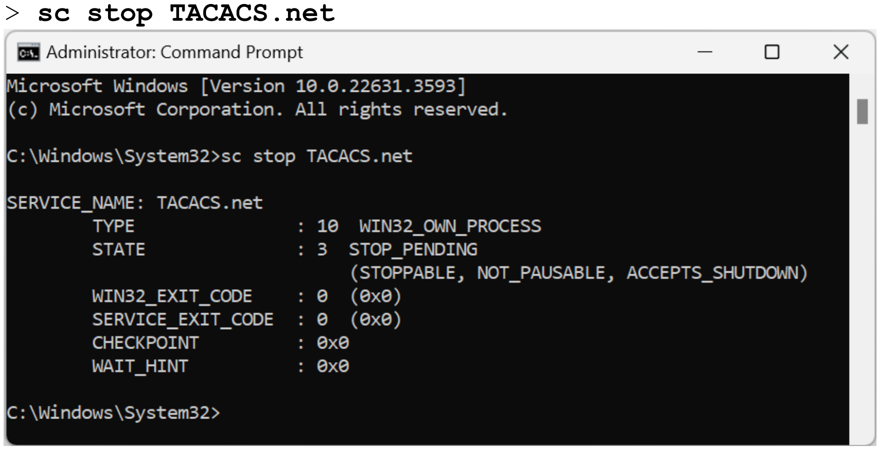


2.8 In addition, open the **authentication.xml** file, uncomment <Users> under the <UserGroup> and modify the user name and login password of your choice:



2.9 When done, restart the TACACS+ service by entering:

```
> sc stop TACACS.net
```



And then

```
> sc start TACACS.net
```

2.10 You may verify that your TACACS.net service is running at IP address and port number as configured with the command:

```
> netstat -an
```

2.11 Configure the switch as TACACS+ client by following the configuration in Lab 3.2 Notes pg 11.

Note: Ensure that both TACACS+ client (switch) and server (TACACS+ PC) are having the same shared secret password by configuring the command key with the shared secret password you have entered in Step 2.4.

WARNING: Do NOT save your configuration to the running-config so that if you make any mistake and are unable to get authentication to work, you are able to simply reload or power down/up the network device to try again.

2.12 When done, enter 'exit' (multiple times) until you are disconnected.

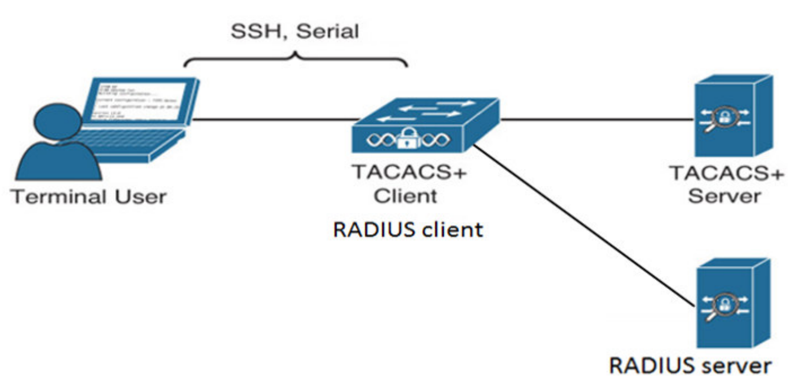
2.13 Try to enter into CLI user mode using the console cable again. Are you able to login using the username and password that you've configured in the TACACS+ server?

2.14 Similarly, try using SSH on Tera Term, are you able to login using the username and password that you've configured in the TACACS+ server?

2.15 If you are unable to login, try running Wireshark at TACACS+ server to troubleshoot whether authentication request/reply messages are being received/sent as shown in Lab 3.2 Notes pg 18. In the worst case, reload and restart your configuration to try again.

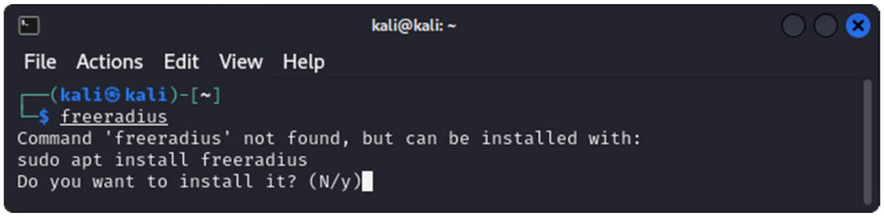
PART 3: Configuring Authentication using Centralised RADIUS Server

3.1 Extend your network topology in Part 2 by connecting a third PC to the switch to function as the RADIUS server as shown:



3.2 Boot up the RADIUS PC in Kali Linux and configure suitable IP address so that it can ping the switch also.

3.3 Start a terminal to install freeRADIUS as shown:



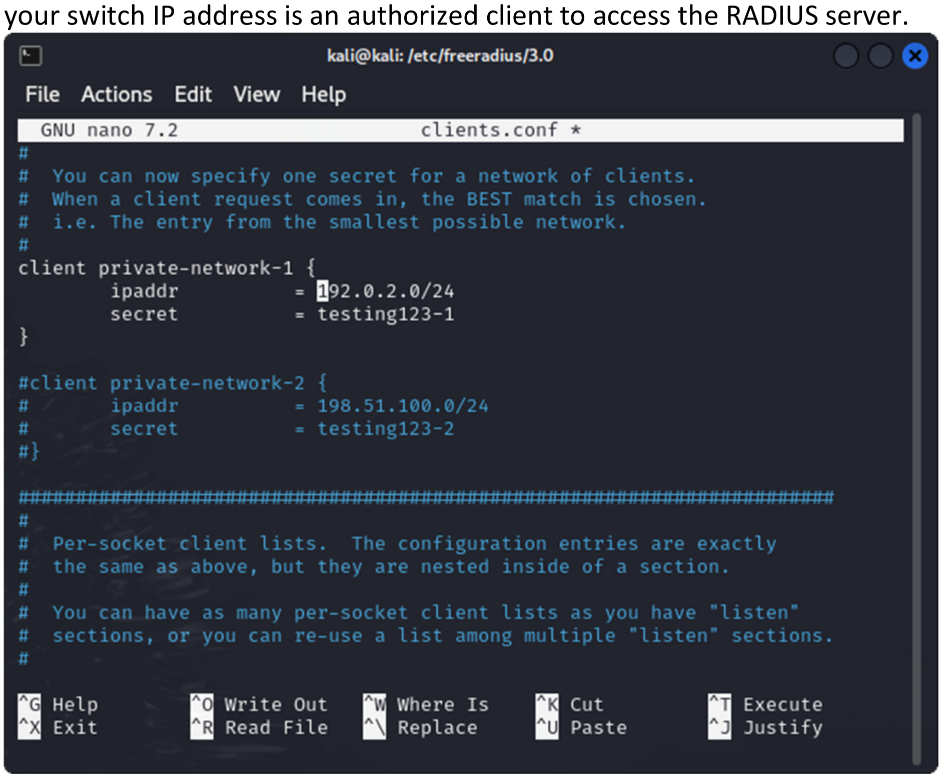
3.4 After installation, the configuration files can be found at `>/etc/freeradius/3.0`. To access it, change the directory permission using the **chmod** command:

```
(kali@kali)-[~]
$ sudo chmod o+rx /etc/freeradius
$ cd /etc/freeradius/3.0
```

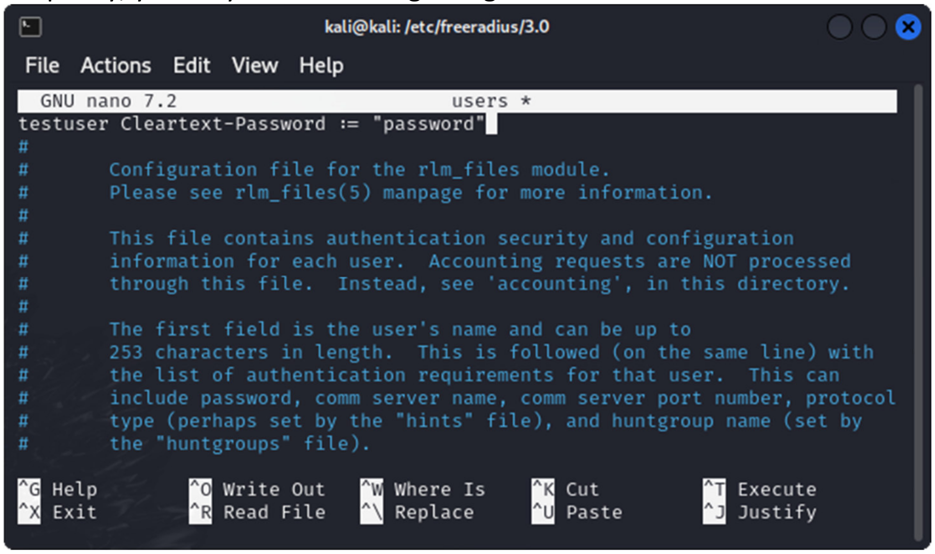
3.5 Open the **clients.conf** using any text editor, say nano editor:

```
(kali@kali)-[/etc/freeradius/3.0]
$ sudo nano clients.conf
```

3.6 Uncomment private-network-1 and update the IP address block and common shared secret between the RADIUS client (switch) and RADIUS server (Kali) to ensure that your switch IP address is an authorized client to access the RADIUS server.



3.7 Similarly, open the **users** file to insert user name and password of your choice. For simplicity, you may add at the beginning of the file:



3.8 When ready, run RADIUS server:

```
(kali@kali)-[/etc/freeradius/3.0]
$ sudo freeradius -X
```

3.9 Start another terminal on RADIUS PC and run **radtest** locally to verify that the server is operating correctly.

3.10 Next, configure the switch as RADIUS client in addition to TACACS+ as shown in Lab 3.2 Notes pg 15.

Note: Ensure that both RADIUS client (switch) and server (RADIUS PC) are having the same shared secret password by configuring the command key with the shared secret password you have entered in Step 3.6.

WARNING: Do NOT save your configuration to the running-config so that if you make any mistake and are unable to get authentication to work, you are able to simply reload or power down/up the network device to try again.

3.11 For experiment, configure different usernames and passwords for TACACS+ server, RADIUS server and local login.

3.12 When done, enter 'exit' (may need multiple times) until you are disconnected.

3.13 Now try to enter into CLI user mode using console cable again. Which username and password are you able to use to login?

3.14 Simulate TACACS+ server is down by disconnecting it from the switch, then try login again. Which username and password are you able to use to login?

3.15 Next, simulate RADIUS server is down by disconnecting it from the switch also, then try login again. Which username and password are you able to use to login?

3.16 If you are unable to login, try running Wireshark at TACACS+ and RADIUS servers to troubleshoot whether authentication request/reply messages are being received/sent as shown in Lab 3.2 Notes pg 18 – 19. In the worst case, reload and restart your configuration to try again.